

Neodymium

60
Nd

When mixed with iron and boron, neodymium makes the most powerful magnets in the world, which can lift objects up to 1,000 times their own weight. They are known as NIB magnets.

ATOMIC MASS 144.242

STATE Solid

DISCOVERY 1885 (Carl Auer von Welsbach)

Laboratory
sample of pure
neodymium



Promethium

61
Pm

Every form of this lanthanide is highly radioactive. Any natural promethium that was in Earth's rocks broke up billions of years ago. However, new promethium atoms are constantly being formed when the atoms of heavier elements break apart – only for the new promethium atoms to also split up. Scientists can make small amounts of this element artificially.

ATOMIC MASS (145)

STATE Solid

DISCOVERY 1945 (Jacob A Marinsky, Lawrence E Glendenin, and Charles D Coryell)

Europium

Although it is named after the continent of Europe, europium is mainly found in rocks in American and Chinese mines. Europium compounds glow red in ultraviolet light, so they are added to bank notes, including euro notes, to prove they are authentic.

Laboratory sample
of pure europium

